

HUAWEI CONNECT: THE FUTURE OF 5G & WIFI 6 POWERING AN INTELLIGENT WORLD



WEDNESDAY, 12 FEBRUARY, 2025

GROUP 6

HUAWEI - LEADING THE ICT REVOLUTION

HUAWEI, a global leader in ICT infrastructure and smart devices, operates in over 170 countries. With a commitment to innovation, the company has pioneered advancements in AI chips, 5G networks and WiFi 6 connectivity, shaping the vision of an intelligent, connected world. In Malaysia, Huawei stands as the largest ICT solution provider, ranking among the top 5 employer brands and winning the Cybersecurity Innovation Award in 2020.



5G - SPEED BEYOND IMAGINATION

What used to take hours is now done in seconds.

The fifth-generation (5G) wireless technology revolutionizes communication with:

- Ultra-low latency: 1 ms (millisecond)
- High-speed data rate: Up to 10Gbps peak speed
- Massive connectivity: 1 million devices/km²

Innovative 5G Applications by Huawei:

- Smart & Virtual Tourism: Tourists can live-stream destinations in real time.
- 5G Industrial Parks: Supporting hi-tech industries and creating new job opportunities.
- Robotic Integration: Increasing productivity by 30% through automation.
- AR/VR Education: Transforming learning with interactive, real-world applications.

With urban coverage ranging from 700-3025Mbps, 5G enables seamless experiences for:

- AR & VR: Enhancing immersive gaming, virtual tourism, and interactive education.
- Automation: Boosting efficiency in manufacturing, transportation, and mining.
- 5G Live Broadcast & Private Networks: Elevating media and business communications.
- Smart Cities: Revolutionizing healthcare, education, and industrial operations



Wi-Fi 6 : Next-Gen Connectivity

Officially certified in 2019, Wi-Fi 6 is based on the 802.11ax standard and offers significant improvements in speed, efficiency, and connectivity. Key features include:

- Blazing Speeds – Up to 9.6 Gbps peak data rates.
- Dual-Band Operation – Works on both 2.4 GHz and 5 GHz spectrums.
- Enhanced Efficiency – Handles 100+ devices simultaneously in high-density environments.
- Lower Latency & Power Consumption – 30% reduction in energy usage for connected devices.
- Stronger Signal & Coverage – Huawei's 8T8R spatial streams increase transmission rates and improve bandwidth by 2.8 times.

How Does Wi-Fi 6 Improve Bandwidth & Coverage?

Huawei optimizes Wi-Fi 6 by offering:

- Larger Spatial Streams – 2x the previous version (Wi-Fi 5). Higher Transmission Efficiency – 5% increase in subcarrier quantity, 6% longer symbol duration, and 25% more efficient coding, reducing data transfer time significantly.
- Enhanced Signal Quality – SNR (Signal-to-Noise Ratio) boosted by 9dB+ using OFDMA-based RU narrowband transmission.
- Extended Coverage – Huawei's RF optimization and TX power improvements ensure stable 193.9 Mbps speeds over wider areas.

Wi-Fi 6: The Future of Video & Cloud Services

Wi-Fi 6 enhances modern digital experiences with:

- OFDMA Technology – Supports multi-user sequential scheduling for 10x better efficiency.
- BSS Coloring – Allows multiple devices to transmit on the same channel simultaneously, reducing congestion.
- Wide DMA for Low Latency – Cloud VR, multi-screen IPTV, HD online teaching, and e-sports now run smoother than ever.

Core Wi-Fi 6 Technologies Impacting Huawei's Future

- 1024-QAM & 8x8 MU-MIMO – Providing 4x larger bandwidth than Wi-Fi 5.
- UL/DL OFDMA & UL/DL MU-MIMO – Supporting 4x higher concurrency than Wi-Fi 5.
- Low Latency & Power Efficiency – 50% lower latency (as low as 20ms) and reduced power consumption.

REFLECTION:

ZARIFF AZAM A24CS0217:

Through this talk, I was exposed to the advances being made in technology, specifically in 5G and Wi-Fi 6, and how they will be implemented in the future. This gives me a better idea on what to focus on to be able to effectively reap the most advantage out of 5G and Wi-Fi 6 and use it to grow in my career. Overall, this talk opened my eyes on how technology can shape a intelligent, more connected world that is designed to make our everyday lives easier.

FATIN SYAKIRAH A24CS0073:

5G and Wi-Fi 6 is the future. This talk answered every question I had about what they are and how they will be implemented in the coming years. 5G and Wi-Fi 6 will bridge the gap between continents and allow us to be more connected with each other. Overall, this talk allowed me to see how we will one day integrate 5G and Wi-Fi 6 into our lives to make this world a smarter, more connected, and everyday life more convenient.

SARVESWARI VASUTHEVAN A24CS0185:

This talk provided insights into how 5G and Wi-Fi 6 will revolutionize connectivity with faster speeds and improved reliability. I learned how these technologies will enhance industries like healthcare, education, making communication more efficient. They will bridge gaps between communities, enable real-time data processing, and create a smarter, more connected world. Overall, this talk helped me understand how 5G and Wi-Fi 6 will shape the future, making everyday life more convenient and seamless.

NUREEN FATINI A24CS0169:

After attending this talk, I got more exposure to what behind new Technologies and why they are much more fast and have smooth transition. From the foundation of 5G and Wi-Fi 6 I got to include that these are the starter of many more technology that can be create in the future. From this, I understand I need to learn more about them in order to create better life with better technology. I also need to strengthen my base knowledge about network to understand more about these technology

WONG ZI NING A24CS0313:

This talk gave me a better understanding of how 5G and Wi-Fi 6 are changing the way we connect and communicate. 5G improves outdoor connectivity, making smart cities and remote work more efficient. Meanwhile, Wi-Fi 6 enhances indoor experiences, providing faster speeds and better connections in homes, schools and offices. These innovations make life easier by reducing internet lag, improving communication and supporting new technologies like AR/VR and smart healthcare. Overall, this talk showed me how technology is shaping a smarter, more connected world and making everyday life more convenient.



Huawei Technologies

Wi-Fi 6

 Huawei AirEngine

- Highest performance
- Most stable experience
- Most comprehensive IoT applications
- Most secure
- 2x speed vs industry Wi-Fi 6

5G

- Smart Antenna – Improves SNR, expanding coverage radius by 20%.
- Dynamic Turbo – Ensures low latency for mission-critical services.
- Network Distribution – Adjusts coverage based on user density.
- Lossless Roaming – Achieves zero packet loss in service roaming.

Wi-Fi 6 vs. 5G: Where They Shine

Wi-Fi 6 is ideal for indoor environments, providing high-speed connectivity in homes, malls, universities, classrooms, and stadiums. It excels in ultra-high-density areas with multiple connected devices.

On the other hand, 5G is designed for outdoor coverage, delivering fast internet across cities, highways, plazas, and scenic areas. It ensures seamless connectivity in urban settings and wide outdoor spaces.